

LAB ASSIGNMENT – 4.5

NAME – SHREE PRIYA

2303A51465

BATCH – 29

Objective: To explore and compare Zero-shot, One-shot, and Few-shot prompting techniques for classifying emails into predefined categories using a large language model (LLM).

1. Suppose that you work for a company that receives hundreds of customer emails daily. Management wants to automatically classify emails into categories like "Billing", "Technical Support", "Feedback", and "Others" before assigning them to appropriate departments.

Instead of training a new model, your task is to use prompt engineering techniques with an existing LLM to handle the classification.

Tasks to be completed are as below

a. Prepare Sample Data:

- Create or collect 10 short email samples, each belonging to one of the 4 categories.

b. Zero-shot Prompting:

- Design a prompt that asks the LLM to classify a single email without providing any examples.

- Example prompt:

“Classify the following email into one of the following categories:

Billing, Technical Support, Feedback, Others. Email: ‘I have not received my invoice for last month.’”

c. One-shot Prompting:

- Add one labeled example before asking the model to classify a new email.

d. Few-shot Prompting:

- Use 3–5 labeled examples in your prompt before asking the model to classify a new email.

e. Evaluation:

- Run all three techniques on the same set of 5 test emails.
- Compare and document the accuracy and clarity of responses.

```
# 1. EMAIL CLASSIFICATION

def email_zero_shot(email):
    prompt = f"""
    Classify the email into:
    Billing, Technical Support, Feedback, Others

    Email: {email}
    Category:
    """
    return llm_simulator(prompt)

def email_one_shot(email):
    prompt = f"""
    Example:
    Email: My payment was deducted twice.
    Category: Billing

    Now classify:
    Email: {email}
    Category:
    """
    return llm_simulator(prompt)

def email_few_shot(email):
    prompt = f"""
    Example 1:
    Email: I was charged extra.
    Category: Billing
    Example 2:
    Email: App crashes on login.
    Category: Technical Support
    Example 3:
    Email: Great service!
    Category: Feedback
    Now classify:
    Email: {email}
    Category:
    """
    return llm_simulator(prompt)
```

Travel Query Classification

Scenario:

A travel assistant must classify queries into Flight Booking, Hotel Booking, Cancellation, or General Travel Info.

Tasks:

- a. Prepare labeled travel queries.
- b. Apply Zero-shot prompting.
- c. Apply One-shot prompting.
- d. Apply Few-shot prompting.
- e. Compare response consistency

2. TRAVEL QUERY CLASSIFICATION

```
def travel_few_shot(query):  
    | prompt = f"""  
    Example 1:  
    Query: Book a flight to Delhi  
    Category: Flight Booking  
  
    Example 2:  
    Query: Cancel my hotel booking  
    Category: Cancellation  
  
    Example 3:  
    Query: Best time to visit Goa?  
    Category: General Travel Info  
  
    Now classify:  
    Query: {query}  
    Category:  
    """  
    | return llm_simulator(prompt)
```

Programming Question Type Identification

Scenario:

A coding help chatbot must classify queries into Syntax Error, Logic Error, Optimization, or Conceptual Question.

Tasks:

- Prepare coding-related user queries.
- Perform Zero-shot classification.
- Perform One-shot classification.
- Perform Few-shot classification.
- Analyze improvements in technical accuracy.

3. PROGRAMMING QUESTION CLASSIFICATION

```
def programming_few_shot(question):  
    prompt = f"""  
    Example 1:  
    Question: Missing semicolon error  
    Category: Syntax Error  
  
    Example 2:  
    Question: Output is wrong  
    Category: Logic Error  
  
    Example 3:  
    Question: How to reduce time complexity?  
    Category: Optimization  
  
    Now classify:  
    Question: {question}  
    Category:  
    ""  
    return llm_simulator(prompt)
```

Social Media Post Categorization

Scenario:

A social media analytics tool must classify posts into Promotion, Complaint, Appreciation, or Inquiry.

Tasks:

1. Prepare sample social media posts.
2. Use Zero-shot prompting.
3. Use One-shot prompting.
4. Use Few-shot prompting.
5. Analyze informal language handling.

```

# 4. SOCIAL MEDIA POST CLASSIFICATION

def social_few_shot(post):
    prompt = f"""
Example 1:
Post: Huge sale today!
Category: Promotion

Example 2:
Post: Worst service ever
Category: Complaint

Example 3:
Post: Love this brand
Category: Appreciation

Now classify:
Post: {post}
Category:
"""
    return llm_simulator(prompt)]

# MAIN EXECUTION

print("\n===== EMAIL CLASSIFICATION =====")
print(email_zero_shot("I have not received my invoice"))
print(email_one_shot("Unable to reset my password"))
print(email_few_shot("The website is very slow"))

print("\n===== TRAVEL QUERY =====")
print(travel_few_shot("Cancel my flight ticket"))

print("\n===== PROGRAMMING QUESTION =====")
print(programming_few_shot("why is my loop running infinitely?"))

print("\n===== SOCIAL MEDIA POST =====")
print(social_few_shot("Do you ship internationally?"))

```

OUTPUT:

===== PROGRAMMING QUESTION =====

--- PROMPT SENT TO MODEL ---

Example 1:

Question: Missing semicolon error

Category: Syntax Error

Example 2:

Question: Output is wrong

Category: Logic Error

Example 3:

Question: How to reduce time complexity?

Category: Optimization

Now classify:

Question: Why is my loop running infinitely?

Category:

Predicted Category

===== SOCIAL MEDIA POST =====

--- PROMPT SENT TO MODEL ---

Example 1:

Post: Huge sale today!

Category: Promotion

Example 2:

Post: Worst service ever

Category: Complaint

Category: Promotion

Example 2:

Post: Worst service ever

Category: Complaint

Example 2:

Post: Worst service ever

Category: Complaint

Example 2:

Post: Worst service ever

Category: Complaint

Post: Worst service ever

Category: Complaint

Example 3:

Example 3:

Post: Love this brand

Category: Appreciation

Now classify:

Post: Love this brand

Category: Appreciation

Now classify:

Category: Appreciation

Now classify:

Post: Do you ship internationally?

Now classify:

Post: Do you ship internationally?

Post: Do you ship internationally?

Category: